

KUSHAGRA SINGH TIWARI

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PROFESSIONAL SUMMARY

Data Engineer / Python Developer with **2+ years** of experience building scalable **ETL/ELT pipelines**, automating data workflows, and developing AI-powered applications. Experienced in **Python, SQL, Databricks, Oracle, PostgreSQL, PySpark**, and Generative AI technologies including **LangChain, RAG**, and **vector databases**. Passionate about combining data engineering with AI/ML to build intelligent, production-ready data solutions.

CORE SKILLS

Programming & Scripting: Python, SQL, PL/SQL, Shell Scripting

Data Engineering & Big Data: ETL/ELT, Apache Hadoop, Apache Spark, Snowflake, Databricks, Hive, PySpark, Data Pipelines, Data Processing, Data Integration, Data Validation, REST APIs, CI/CD, Terraform

GenAI & AI: LangChain, Ollama, Chroma, RAG, Vector Embeddings, LangGraph, CrewAI, MCP, Machine Learning, Deep Learning

Databases, File Formats & Cloud: Oracle, PostgreSQL, MySQL, SQLite, JSON, CSV, Parquet, AWS, GCP

Tools & Technologies: Git, Linux/UNIX, Pandas, Power BI, Postman, SFTP Automation

EXPERIENCE

Splash Business Intelligence Pvt. Ltd. | Associate Software Engineer Feb 2024 – Present | Hyderabad

- Optimized enterprise data workflows by developing **high-performance PL/SQL procedures, functions, and views**, improving data processing efficiency by **25%** across **Oracle Cloud, UKG**, and **SAP** systems.
- Designed and maintained ETL pipelines using **PL/SQL, REST APIs, and Postgres databases** to ingest, transform, and load currency exchange data for global clients.
- Developed Python-based file ingestion and **SFTP automation workflows** for enterprise ETL pipelines, reducing manual processing effort by **90%**.
- Developed **PySpark-based validation frameworks** in **Databricks** to verify incremental ETL loads, ensuring data consistency and accuracy across **Bronze, Silver**, and **Gold layers** for multiple enterprise modules.
- Implemented a **centralized telemetry automation pipeline** for **40+ clients**, consolidating ETL execution metrics into a single dataset, loading it into a database, and enabling **automated email alerts and reporting**.

PHN Technology Pvt. Ltd. | Programming Intern 04/2023 – 06/2023 | Pune

- Developed a **Python-based machine learning model** using **NumPy, Pandas, Scikit-learn, Matplotlib, and OpenPyXL** to predict linear relationships from structured datasets.
- Built and optimized a **data preprocessing and feature engineering pipeline**, improving model accuracy by **~15%**.

PROJECTS

Local AI Agent (RAG-Based Intelligent Q&A System) Nov 2025 – Dec 2025

- Built a **locally deployed AI agent** using **Retrieval-Augmented Generation (RAG)** for contextual document-based Q&A.
- Integrated **Ollama** to run LLMs locally, enabling **offline and private inference** without external APIs.
- Implemented a **Chroma-based vector search pipeline** for fast semantic document retrieval.
- Orchestrated LLM workflows using **LangChain** and optimized document preprocessing with **Pandas**.
- Tech Stack:** Python, LangChain, Ollama, FAISS, Pandas

Image Forgery Detection & Localization Aug 2023 – Dec 2023

- Developed a **U-Net-based 2D CNN model** to detect copy-move and splicing image forgeries.
- Performed **image preprocessing, annotation handling, and dataset preparation** to improve model robustness and generalization. Achieved **76% Binary IoU**, demonstrating strong localization performance.
- Tech Stack:** Python, Tensorflow, Keras

EDUCATION

Master of Computer Applications (MCA) | University of Hyderabad, Hyderabad | CGPA: 7.4 2022 - 2024

Bachelor of Science | APS University, Rewa, MP | Percentage: 69.15% 2018 - 2021